

New and Old ideas in Macro reading group

Inflation is Conflict (WP NBER 2023)

Wage-Price Spirals (BPEA Fall 2023)

Lorenzoni and Werning

Judith Kleman (ESCP / Crest)

11th March 2026

Context

Recent Inflation and the Central Bank Trade-off

High inflation after the pandemic

- Inflation surged in the euro area between **2021 and 2023**
- Initial drivers largely related to **supply shocks**:
 - Surge in energy prices
 - Supply chain disruptions
- **Goods prices increased rapidly**, while **wages adjusted more slowly**
- As a result, **real wages declined** in most advanced economies

A monetary policy dilemma

CB faced a **trade-off**:

- **Tolerate higher inflation temporarily**
 - Facilitates the adjustment of **real wages** after a supply shock
 - avoids large recessionary costs
- **Tighten monetary policy**
 - Bring inflation down quickly
 - Prevent the emergence of a **wage-price spiral**

ECB response: rapid monetary tightening from **2022** to prevent persistent inflation

Highly criticized: An appeal to understand clearly the potentiality of WP spiral and the link with real wage dynamics

Are Standard Inflation Theories Sufficient?

Monetary view :

- Inflation arises from *“too much money chasing too few goods”*
- Focus on money supply

NK view : Blanchard (1986), Blanchard & Galí (2007), Erceg et al. (2000)

- Inflation emerges from **nominal rigidities** and economic activity
- Higher demand or costs → higher marginal costs → higher prices
- Monetary policy stabilizes inflation by stabilizing the **output gap**

A common limitation

- These frameworks say fw things about the **mechanism that generates sustained inflation**
- In particular, they do not isolate the role of **conflicts over relative prices** between economic agents

Alternative tradition: inflation as distributive conflict (Kalecki (1971), Rowthorn (1977), Marglin & Bhaduri (1990))

- Inflation acts as an adjustment mechanism for the distribution of income
- But remained outside mainstream macro

Problematics

- Isolate the conflictual part of inflation
- Show how it operates in standard NK models, and could account for sustained inflation

Two papers of Lorenzoni and Werning

- "Inflation is Conflict" (2023):
 - Isolating the **conflict**'s part of inflation with a highly stylized dynamic model with staggered prices
 - Showing that in the version with random matching, forward looking inflation will emerge, and may amplify or louder the conflict mechanism.
 - Decompose inflation in two parts (**conflict and adjustment**) in stylized wage price model with **adjustment frictions**
- "Wage-Price Spirals" (2023):
 - Shows how this decomposition appears in a **standard New Keynesian model** with sticky prices and wages
 - Studies the dynamics of **price inflation, wage inflation and real wages** after demand and supply shocks
 - Characterizes the implications for the **wage-price spiral** and for **Optimal Monetary OPolicy**

ooooo



ooooo
oo



oo
ooo

oo
oo

ooooo

ooo

oooo

The decomposition of the inflation mechanism

Model Setup: Inflation as pure conflict

Goal: Isolate inflation generated purely by *conflict* (disagreement over relative prices), not by money or unemployment.

Environment

- Two agents A, B and two goods A, B .
- Endowments: each period, A has 1 unit of good A , B has 1 unit of good B .
- Goods are perishable $\Rightarrow$ no storage, no saving, no capital.
- $u(c, c')$

No money / no credit

- Trade occurs via **barter**

Staggered price setting (2-period rigidity)

- Even t : A resets its price $P_t^* = P_t^A = P_{t+1}^A$.
- Odd $t + 1$: B resets its price $P_{t+1}^* = P_{t+1}^B = P_{t+2}^B$.
- Terms of trade use the ratio of the last two posted prices (**relative price p^***).

Trading protocol

- The agent who *did not* reset price makes a take-it-or-leave-it offer (the “buyer”).
- If rejected, each consumes own endowment.

Model of inflation as conflict : Equilibrium Inflation

Step 1: Buyer's problem

The buyer chooses consumption given the relative price p :

$$V(p) = \max_{c, c'} u(c, c') \quad \text{s.t.} \quad c + pc' = 1$$

Optimal demand:

$$c' = D(p)$$

Step 2: Seller's problem

The seller sets the relative price p to maximize utility:

$$p^* = \arg \max_p v(p)$$

where

$$v(p) = u(1 - D(p), pD(p))$$

Optimal price

First-order condition:

$$p^* = \frac{\epsilon(p^*)}{\epsilon(p^*) - 1} \frac{u_c}{u_{c'}}$$

- Sellers behave like **monopolists**: have a mark-up

Implication for inflation (constant and positiv)

$$\frac{P_{t+1}^*}{P_t^*} = p^* > 1$$

Extension: Random Matching and Inflation Expectations

- In the last model, agents permanently paired
- Static price problem: the price today has no impact on the relative price tomorrow
- Continuum of agents of each type.
- Each period, each sellers are matched with a **random buyer**
- Prices are set **before knowing the trading partner**.
- The price chosen today affects the **future terms of trade**:
- The seller chooses p with an arbitrage:
 - If anticipated inflation is high, future prices will be higher: agents anticipate unfavourable terms of trade when they are buyers.
- → Inflation depends now on expectations

ooooo



●ooooo
oo



ooooo

ooo

oooo

Inflation mechanism accounting Wages and Prices

Inflation Mechanism: Prices, Wages and Aspirations

Environment

- Continuum of firms and workers.
- Staggered price and wage setting (Calvo):
 - Firms reset prices with probability λ_p
 - Workers reset wages with probability λ_w
- Inflation results from gradual adjustment of prices and wages

$$\pi_t = \lambda_p(p_t^* - p_t), \quad \pi_t^w = \lambda_w(w_t^* - w_t)$$

Aspirations

- Firms target a desired markup: $f_t = p_t^* - w_t$
- Workers target a desired real wage: $g_t = w_t^* - p_t$

Conflict

- f_t and g_t capture firms' and workers' aspirations (let consider them as exogenous for the moment).
- Conflict arises when $f_t < g_t$.

Resulting inflation dynamics: based on the fact that the real wage is $\omega_t = w_t - p_t$

$$\pi_t = \lambda_p(\omega_t - f_t), \quad \pi_t^w = \lambda_w(g_t - \omega_t)$$

- Price and wage inflation reflect the gap between the current real wage ω_t and each side's desired level.

Reference real wage

- In case of conflict, since both aspiration cannot be attained at the same time
- the real wage will converge toward a weighted average of both aspirations:

$$\tilde{w}_t = \alpha f_t + (1 - \alpha)g_t, \quad \alpha = \frac{\lambda_p}{\lambda_p + \lambda_w}$$

Conflict vs Adjustment Inflation

Conflict inflation:

$$\Pi_t^C = (\lambda_p + \lambda_w)\alpha(1 - \alpha)(g_t - f_t)$$

- Conflict inflation is proportional $g_t - f_t$, which can generate inflation, or deflation
- $\alpha(1 - \alpha)$: intensity of conflict transmission
- $(\lambda_p + \lambda_w)$: global speed of nominal adjustment

Adjustment inflation: Drives the adjustment of the real wage: $\dot{\omega}_t = \Pi_t^A$

$$\Pi_t^A = (\lambda_p + \lambda_w)(\tilde{\omega}_t - \omega_t)$$

- Arises when the real wage differs from its reference wage, depending on the speed of adjustment

Proposition 2: Inflation decomposition

$$\pi_t = \Pi_t^C - \alpha\Pi_t^A, \quad \pi_t^w = \Pi_t^C + (1 - \alpha)\Pi_t^A$$

Proposition 3: Price and Wage Inflation Requires Conflict

$$\Pi_t^C = \alpha \pi_t^w + (1 - \alpha) \pi_t$$

- If price and wage inflation are both positive:

$$\pi_t > 0, \quad \pi_t^w > 0$$

then

$$\Pi_t^C > 0$$

- Thus, **simultaneous price and wage inflation requires** $g_t > f_t$

Proposition 4: Long-Run Inflation is Driven by Conflict

Define long-run average inflation rates

$$\bar{\pi} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \pi_t dt \quad \bar{\pi}^w = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \pi_t^w dt \quad \overline{\Pi^C} = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T \Pi_t^C dt$$

Result ¹

$$\bar{\pi} = \bar{\pi}^w = \overline{\Pi^C}$$

- The **adjustment component** only moves the real wage and therefore cannot sustain inflation over long periods.
- Adjustment forces (Π_t^A) generate temporary inflation while the real wage converges to its target.

¹Integrating the identity $\dot{w}_t = \pi_t^w - \pi_t$ over time and dividing by T implies that the difference between average wage and price inflation vanishes as $T \rightarrow \infty$, so that $\bar{\pi} = \bar{\pi}^w$. Combining this result with Proposition 3, $\Pi_t^C = \alpha \pi_t^w + (1 - \alpha) \pi_t$, yields

Forward-Looking Aspirations

- Firms and workers reset prices and wages only occasionally (Calvo).
- When setting them, they must anticipate **future inflation** :

$$p_t^* = (\rho + \lambda_p) E_t^p \int_0^\infty e^{-(\rho + \lambda_p)s} (w_{t+s} - \phi_{t+s}) ds$$

$$w_t^* = (\rho + \lambda_w) E_t^w \int_0^\infty e^{-(\rho + \lambda_w)s} (p_{t+s} + \gamma_{t+s}) ds$$

P depends on expected future wages / W on expected future prices : W-P spiral

Target VS aspiration

- ϕ, γ : **(structural) target margins and real wages**
- f, g : **effective aspirations, adjusted by expected inflation** (endogenized)

At the steady state:

$$g - f = \gamma - \phi + \frac{\hat{\pi}^w}{\rho + \lambda_p} + \frac{\hat{\pi}^p}{\rho + \lambda_w}$$

- $\hat{\pi}^p, \hat{\pi}^w$: expected price and wage inflation at the S.S.
- Expected inflation may ↗ the gap between worker and firm aspirations

Forward-looking Inflation and Distributive Conflict

- assumption that firms and workers share RE with no aggregate uncertainty
- Inflation becomes **forward-looking**

Inflation dynamics

$$\pi_t = \Lambda_p \int_0^\infty e^{-\rho s} (\omega_{t+s} - \phi_{t+s}) ds \quad \pi_t^w = \Lambda_w \int_0^\infty e^{-\rho s} (\gamma_{t+s} - \omega_{t+s}) ds$$

with $\Lambda_p = \lambda(\rho + \lambda_p)$ and $\Lambda_w = \lambda_w(\rho + \lambda_w)$, the PC coefficients

- Inflation depends on the **present value of future distributive gaps**

Proposition 5: Forward-looking decomposition

$$\pi_t = \Pi_t^{C,f} - \alpha^f \Pi_t^{A,f} \quad \pi_t^w = \Pi_t^{C,f} + (1 - \alpha^f) \Pi_t^{A,f}$$

and then

$$\Pi_t^{C,f} = \alpha \pi_t^w + (1 - \alpha) \pi_t$$

Even with RA, **persistent** π & π^w **requires distributive conflict** between targets.

How this mechanism appears in a NK DSGE

Model Setup

Environment: New Keynesian model with wage and price rigidities

Households

- Preferences over consumption and labor:

$$\max \int_0^{\infty} e^{-\rho t} \left(\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{\Phi_t}{1+\eta} N_t^{1+\eta} \right) dt$$

- Supply differentiated labor varieties through unions
- C_t is an aggregate of a continuum of varieties of goods following Dixit-Stiglitz

Firms

- Monopolistic producers with production function

$$Y_{jt} = F(L_{jt}, X_{jt}) = \left(a_L L_{jt}^{\frac{\epsilon-1}{\epsilon}} + a_X X_{jt}^{\frac{\epsilon-1}{\epsilon}} \right)^{\frac{\epsilon}{\epsilon-1}}$$

- X_{jt} : non-labor input (e.g. energy)
- Set prices subject to Calvo rigidity (λ_p)
- When allowed to reset, choose P_t^* to maximize discounted profits

Unions (wage setters)

- Supply differentiated labor types
- Set nominal wages subject to Calvo rigidity (λ_w)
- When resetting, choose W_t^* to maximize households' utility

Price and Wage Setting

By solving the maximisations problems of the firms and of the HH

Optimal reset price

Firms choose a price that is a markup over expected future marginal costs:

$$p_t^* = (\rho + \lambda_p) \int_t^\infty e^{-(\rho + \lambda_p)(\tau - t)} (w_\tau - mpl_\tau) d\tau$$

Optimal reset wage

Unions choose wages anticipating future prices:

$$w_t^* = (\rho + \lambda_w) \int_t^\infty e^{-(\rho + \lambda_w)(\tau - t)} (p_\tau + mrs_\tau) d\tau$$

We can apply the results that they found in the previous article, by considering:

Firms' aspiration $\phi_t \rightarrow mpl_t$
 Workers' aspiration $\gamma_t \rightarrow mrs_t$

Classical Interpretation of the Two Phillips Curves

Standard New Keynesian framework (continuous time)

Price Phillips Curve

$$\rho\pi_t = \Lambda_p mc_t + \dot{\pi}_t$$

where

$$mc_t = \omega_t - mpl_t$$

is the **real marginal cost**.

Wage Phillips Curve

$$\rho\pi_t^w = \Lambda_w (mrs_t - \omega_t) + \dot{\pi}_t^w$$

Standard interpretation

- Firms raise prices when **marginal costs** increase
- Workers raise wages when the real wage falls below their **marginal rate of substitution**
- Price Inflation is usually interpreted as being driven by **demand pressures** through marginal costs
- But mct_t can be reinterpreted as conflict over real wage

Reinterpreting the Two Phillips Curves as a Distributive Conflict

Inflation equations²

$$\begin{aligned}\rho\pi_t &= \Lambda_p(\omega_t - mpl_t) + \dot{\pi}_t \\ \rho\pi_t^w &= \Lambda_w(mrs_t - \omega_t) + \dot{\pi}_t^w\end{aligned}$$

Interpretation

- **Firms desired real wage:** $\omega = mpl$ If $\omega_t > mpl_t \Rightarrow$ marginal costs too high $\Rightarrow$ firms raise prices.
- **Workers desired real wage:** $\omega = mrs$ If $\omega_t < mrs_t \Rightarrow$ real wage too low $\Rightarrow$ workers raise wages.
- Inflation arises from a **conflict over the real wage:** $mpl_t < mrs_t$

The real wage is still definit by $\dot{\omega} = \pi_t^w - \pi_t$

Those 3 equations, given the initial condition for ω_0 give the unique paths for price and wage inflation.

²We can take the laws of motion $\dot{p}_t = \lambda_p(p_t^* - p_t)$, $\dot{w}_t = \lambda_w(w_t^* - w_t)$, by combining with optimal prices and wages, which gives

Inflation and Real Wage Dynamics

We can perfectly reapply the the decomposition of Lorenzoni & Werning (2022)

$$\pi_t = \Pi_t^C - \frac{\Lambda_p}{\Lambda_p + \Lambda_w} \Pi_t^A \quad \pi_t^w = \Pi_t^C + \frac{\Lambda_w}{\Lambda_p + \Lambda_w} \Pi_t^A$$

- With

$$\Pi_t^C = \frac{\Lambda_p \Lambda_w}{\Lambda_p + \Lambda_w} \int_t^\infty e^{-\rho(\tau-t)} (mrs_\tau - mpl_\tau) d\tau \quad \Pi_t^A = \dot{\omega}_t$$

- Π_t^C , depends on the discounted sum of future gaps between workers' and firms' real wage aspirations
- Because price and wage setting are forward-looking, current inflation reflects the expected present value of these future conflicts.
- time the strenght of wage price spiral (limited by the most rigid side)

We can reapply the final result of Lorenzoni Werning (2022)

Persistent inflation arises from **distributive conflict**

ooooo

o
o
o

ooooo
oo

ooo
oo
o

oo
oo

oo
oo

ooooo

ooo

oooo

Inflation and real wage

Real Wage Dynamics and Conflict

Combining the price and wage Phillips curves gives the dynamics of the real wage:

$$\ddot{\omega}_t = \rho \dot{\omega}_t + (\Lambda_p + \Lambda_w)(\omega_t - \tilde{\omega}_t)$$

where the **reference real wage** is

$$\tilde{\omega}_t = \alpha \text{mpl}_t + (1 - \alpha) \text{mrs}_t, \quad \alpha = \frac{\Lambda_p}{\Lambda_p + \Lambda_w}$$

- The reference level $\tilde{\omega}_t$ is a weighted average of the firms and workers aspirations, with weights determined by the relative rigidity of prices and wages.
- If $\omega_t > \tilde{\omega}_t \Rightarrow$ price inflation dominates $\Rightarrow$ real wage falls
- If $\omega_t < \tilde{\omega}_t \Rightarrow$ wage inflation dominates $\Rightarrow$ real wage rises

The direction of ω movements does not indicate the strength of the wage–price spiral

This dynamic equation provides the basis for the phase-diagram analysis of real wage adjustment.

Permanent fall in mpl

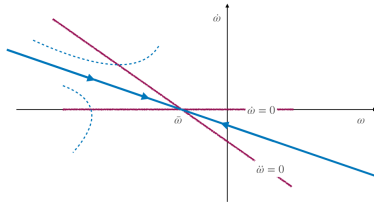


Figure 1: A permanent shock

- Initially, SS with $mpl = mrs$.
- At $t=0$, mpl fall contrary to mrs : $\tilde{\omega} \searrow$
- $\tilde{\omega} < \omega_0$: ω has to fall/
- Since $\dot{\omega} = \pi_t^w - \pi_t$, we need $\pi_t > \pi_t^w$ in order to $\searrow \omega$
- Inflation is therefore initially driven by two forces:
 - a permanently higher *conflict inflation*, which correspond to LT inflation
 - temporary *adjustment inflation* due to the fall in the real wage, which vanishes

Transitory shock in mpl

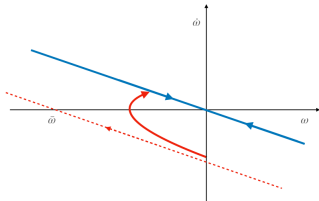


Figure 2: A transitory shock

- Initially, the real wage becomes too high relative to productivity:
 $\dot{\omega}_0 < 0 \Rightarrow \omega_t$ falls.
- Before T , expectations of the future recovery dominate and $\dot{\omega}_t > 0$, so the real wage starts increasing again.
- After T , the shock disappears and the economy converges back to the steady state $\dot{\omega} = 0$.

Implications for inflation

- Initially: **conflict inflation** + **adjustment inflation** $\Rightarrow$ high price inflation.
- As $\dot{\omega}$ turns positive, the adjustment component reverses sign.
- Price inflation can become **negative**, while wage inflation remains positive: Π^C vanishes, and Π^A dominates until ω reaches $\tilde{\omega}$

Persistent shocks in mpl and mrs

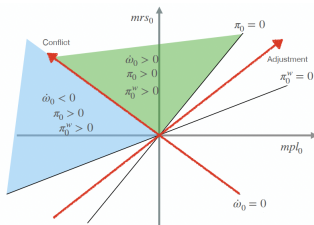


Figure 5: Regions for mpl_0 and mrs_0

- when firms and Workers' aspirations change simultaneously, inflation depends on strongness and direction of the conflict AND adjustment part
- The loci for $\pi_0 = 0$ and $\pi_0^w = 0$ are asymmetric.
- $\pi_0 = 0 \not\Rightarrow \pi_0^w = 0$ and vice versa.
- This asymmetry arises because mpl directly affects price setting, while mrs directly affects wage setting.
- Firms control prices while workers influence wages, with cross-effects operating only indirectly through expectations and real wage dynamics.

Demand and Supply shocks

Demand shock: expansion in employment ($n_t = n_0 e^{-\delta t}$)

Effect on firms' and workers' aspirations

$$mpl_t = -\frac{s_X}{\epsilon} e^{-\delta t} n_0 < 0 \quad mrs_t = (\sigma s_L + \eta) e^{-\delta t} n_0 > 0$$

- firms' side: higher production $\nearrow$ the demand for the scarce input, $\nearrow$ marginal costs and $\searrow$ $mpl_t \Rightarrow$ UP pressure on prices.
- The demand shock $\nearrow e$, $\rightarrow \nearrow$ workers' desired ω through two channels.
 - higher economic activity $\nearrow$ consumption : **income effect**. With σ the inverse intertemporal elasticity of substitution and s_L the labor share, higher consumption $\nearrow$ the marginal value of income $\rightarrow \nearrow$ workers' wage demands.
 - supplying more labor raises the marginal disutility of work, governed by inversed Frisch elasticity. As $e \nearrow$, the marginal rate of substitution between consumption and leisure $\nearrow$, meaning that workers require a higher real wage to supply labor.

$$\rightarrow \pi_t > 0, \quad \pi_t^w > 0$$

The real wage response depends on inflationary pressures on each side

$$\dot{\omega}_t < 0 \quad \text{iff} \quad \Lambda_p \frac{s_X}{\epsilon} > \Lambda_w (\sigma s_L + \eta)$$

Supply shock: reduction in the supply of input X

- The scarcity of the input raises firms' marginal costs

$$mpl_t = \frac{s_X}{\epsilon} e^{-\delta t} x_0 < 0$$

- The fall in output reduces workers' wage demands through an income effect (workers ask less for real wage)

$$mrs_t = \sigma_{s_X} e^{-\delta t} x_0 < 0$$

The effects of a supply shock depend crucially on the MP.

- If the CB stabilizes employment $\rightarrow$ prevents quantities from adjusting, firm face higher MC, while workers' wage demands fall (negative income effect)
 - $\pi > \pi^w > 0$,
 - temporary decline in the real wage.
- if MP allows the economy to move to its natural allocation, employment falls and the adjustment of the real wage occurs through nominal prices and wages moving in opposite directions
 - $\pi^w < 0 < \pi$

Which Optimal Monetary Policy

Optimal Monetary Policy

- We now characterize the **optimal monetary policy response** to a supply shock.
- With price and wage rigidities,

Policy problem

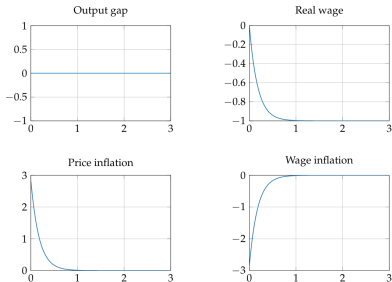
- The central bank maximizes a quadratic approximation of social welfare:

$$\int_0^{\infty} e^{-\rho t} \frac{1}{2} [-(y_t - y_t^*)^2 - \Phi_p \pi_t^2 - \Phi_w (\pi_t^w)^2] dt$$

- Policy chooses the path of activity (output gap) subject to price and wage Phillips curves.
- After a supply shock, the efficient real wage must fall. However, because both prices and wages are sticky, ω can only adjust through movements in price inflation, wage inflation, or economic activity. **This creates a policy trade-off.**
- As a result, the CB must balance these different distortions when guiding the adjustment of the real wage.
- In this environment, the *divine coincidence* disappears³

³In standard New Keynesian models with only price rigidities, stabilizing inflation automatically stabilizes the output gap.

Optimal Policy: Symmetric Case



$$\Phi_p = \Phi_w, \quad \Lambda_p = \Lambda_w$$

The optimal policy splits the real wage adjustment evenly between:

- **price inflation** ($\pi_t > 0$)
- **wage deflation** ($\pi_t^w < 0$)

while keeping

$$y_t - y_t^* = 0$$

Intuition: no reason to favor prices over wages.

Example : Optimal policy with a “hot economy”

$$\Phi_p < \Phi_w, \quad \Lambda_p > \Lambda_w$$

- The supply shock lowers
 - y^{*4}
 - the efficient ω : In both policies, its falls toward its new steady state.

The key difference concerns π^w :

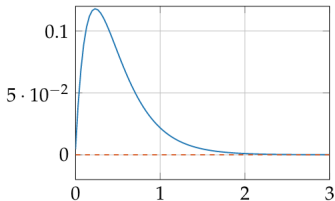
- With a zero-output-gap policy, wage inflation becomes strongly negative.
- Under the optimal policy, stronger activity reduces mpl and increases workers' MRS, raising both price inflation and wage inflation.

Starting from a situation with positive price inflation and negative wage inflation, this **can improve welfare because the welfare cost of wage deflation is larger than the cost of higher price inflation.**

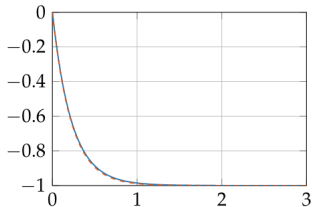
⁴if the CB does not reduce enough y , $y < y^*$

OP with a "hot economy"

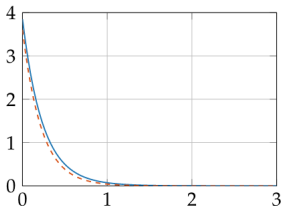
Output gap



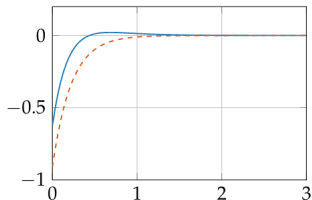
Real wage



Price inflation



Wage inflation



External discussion

External discussion

1. **Come back to the post covid inflation:** ECB's rapid tightening following the inflation surge can be understood as an attempt to prevent a persistent wage–price spiral. However,
 - Tight policy may accelerate the disinflation process through weaker activity, while a more accommodative stance could allow part of the adjustment to occur through higher price inflation and stronger wage growth, especially in case of $\lambda_w \ll \lambda_p$ ⁵
 - The OMP therefore depends on how policymakers weight relative costs.
2. However: The mechanism of inflation led by the conflict of both sides seems discutable: **Recent historical evidence suggests that wage-price spirals are relatively uncommon** (Alvarez et al., WP IMF 2022)
 - In most inflationary episodes, prices tend to increase first, while wages only adjust with a delay. This pattern suggests that the origins of inflation may lie primarily on the firms' side rather than in wage dynamic
 - Seems also that the primary inflation based on supply shocks not enough to explain all of the inflation of the period.

⁵Can be explained by the loss of bargaining power of workers (Ratner and Sims, 2019, FED), and then, a particularly low λ_w

Explore greedflation

Pb in Lorenzoni and Werning analysis and framework:

- Does not take at all into account financial structure
 - no investment (in this specific model)
 - no shareholders payouts
- What is missing in those standard models:
 - Profit can be decomposed into two components: dividends and retained earnings (investment)
 - Dividends usually assume to adjust freely **Evidence from corporate finance shows strong downward rigidity of dividends, and downward rigidity of the ratio profit/corporate payouts**
 - in case of decrease in profit, should be compensated somewhere
- Implication: a three-sided distributive conflict, with **inflation becoming a way to protect shareholders returns.**

Appendix

Solving the Seller's Problem (Monopoly Pricing in Barter)

Step 1: Buyer demand as a function of the relative price $c'(p) = D(p)$

Step 2: Seller's consumption implied by $D(p)$

The seller is endowed with 1 unit of own good. If the buyer purchases $D(p)$ units, the seller:

$$c(p) = 1 - D(p), \quad c'(p) = pD(p)$$

Step 3: Seller chooses the relative price $p^* \in \arg \max_{p>0} v(p) \equiv u(1-D(p), pD(p))$

Step 4: First-order condition

Let $x = D(p)$ and define the (local) elasticity

$$e(p) \equiv -\frac{D'(p)p}{D(p)}.$$

Differentiate $v(p)$:

$$v'(p) = u_c(-D'(p)) + u_{c'}(D(p) + pD'(p)) = 0.$$

Using $D'(p) = -\frac{e(p)}{p}D(p)$, this becomes

$$(e(p) - 1)u_{c'} = \frac{e(p)}{p}u_c \quad \Rightarrow \quad p^* = \frac{e(p^*)}{e(p^*) - 1} \cdot \frac{u_c}{u_{c'}}$$

- The seller sets a **monopoly markup** $\frac{e}{e-1} > 1$ over the MRS $\frac{u_c}{u_{c'}}$.

Proof proposition 2

'Adjustment inflation:

Starting from the definition of adjustment inflation as the change in the real wage,

$$\Pi_t^A \equiv \dot{\omega}_t = \pi_t^w - \pi_t,$$

substituting the inflation equations

$$\pi_t = \lambda_p(\omega_t - f_t), \quad \pi_t^w = \lambda_w(g_t - \omega_t),$$

yields

$$\dot{\omega}_t = (\lambda_p + \lambda_w)(\tilde{\omega}_t - \omega_t),$$

where

$$\tilde{\omega}_t = \alpha f_t + (1 - \alpha)g_t.$$

(conflict inflation):

Conflict inflation is defined as the common component of price and wage inflation that does not affect the real wage:

$$\Pi_t^C \equiv \alpha \pi_t^w + (1 - \alpha)\pi_t.$$

Substituting the inflation equations

$$\pi_t = \lambda_p(\omega_t - f_t), \quad \pi_t^w = \lambda_w(g_t - \omega_t),$$

yields

$$\Pi_t^C = (\lambda_p + \lambda_w)\alpha(1 - \alpha)(g_t - f_t).$$

Deriving the Reference Real Wage

Step 1: Inflation equations

$$\pi_t = \lambda_p(\omega_t - f_t), \quad \pi_t^w = \lambda_w(g_t - \omega_t)$$

with $\omega_t = w_t - p_t$.

Step 2: Look for a weighted combination eliminating ω_t

Consider

$$\Pi_t^C \equiv \alpha \pi_t^w + (1 - \alpha) \pi_t$$

Substituting the two inflation equations gives

$$\begin{aligned} \Pi_t^C &= \alpha \lambda_w (g_t - \omega_t) + (1 - \alpha) \lambda_p (\omega_t - f_t) \\ &= \alpha \lambda_w g_t - (1 - \alpha) \lambda_p f_t + \left[(1 - \alpha) \lambda_p - \alpha \lambda_w \right] \omega_t \end{aligned}$$

Choose

$$\alpha = \frac{\lambda_p}{\lambda_p + \lambda_w}$$

so that

$$(1 - \alpha) \lambda_p = \alpha \lambda_w$$

and the ω_t -term disappears.